domain- or task-specific applications in recommender systems, where the nature of generated text and evaluation criteria may differ substantially. In such settings, personalization—reflecting individual user preferences, tastes, and context—plays a crucial role in determining the perceived quality and relevance of generated content. This contrasts with most general NLP text generation tasks, where personalization is typically less central to evaluation.

3.1.2 Reference-free evaluation. Because collecting human annotations for generative tasks is time-consuming and costly, a growing body of research has explored reference-free evaluation frameworks for tasks requiring an *approximate match evaluation*, [22, 35]. In addition, these metrics are also considered valuable in online scenarios requiring real-time text quality estimation. Formally, when Y is the open-form output space (i.e., any natural language text), a reference-free evaluation metric M can be defined as a function of the input x and the model-generated output $\hat{y}$, which can be viewed as an explicit or implicit conditional generation model [35, 155]:

$$M : X \times Y \to \mathbb{R}, \quad M(x, \hat{y}) = P_\phi(\hat{y} \mid x),$$

where ϕ denotes the parameters of the underlying evaluation model, and higher values of M indicate better-quality outputs. The underlying evaluation model P_ϕ is typically larger or more complex than P_θ, but it also has its own limitations and, therefore, cannot be used to solve the task directly—for instance, it may not be specifically optimized for that task or could be prohibitively expensive to run. Reference-free metrics could employ evaluation models in a zero-shot setup (i.e., pre-trained LLMs such as GPT [48]), while others [127] undergo an optimization process based on some training data to estimate ϕ.

Examples of reference-free metrics, M, are **Prism-src** [155], **COMET-QE** [127], **QuestEval** [141], **ParaScore** [148], **GPTScore** [48], **AlignScore** [183], **ExplicitScore** [23] or **NLI-inspired** metrics [22]. Reference-based metrics for text generation tasks (e.g., **BARTScore**, **BERTScore**) can also be adapted for reference-free evaluation by computing them between the input x and the generated output $\hat{y}$, when x is itself textual [148].

Although reference-free metrics were initially found to underperform compared to reference-based ones, recent work has shown substantial improvements in their ability to approximate human judgments of text quality [47, 148]. *However, their reliability remains inconsistent across different natural language generation tasks, setups, and domains, with correlations to human evaluation varying considerably depending on various specific linguistic attributes (e.g., negation and sentiment polarity are poorly captured by embedding-based metrics in machine translation [47], while lexical divergence in paraphrasing—especially when text is written by humans, is also measured inadequately [148]). This variability highlights an important open research problem: designing generalizable, reference-free evaluation methods that robustly capture human-perceived quality across diverse tasks, data distributions, and complex linguistic phenomena. Addressing this challenge requires a deeper understanding of how subjective dimensions of quality can be operationalized. Furthermore, extending such evaluation metrics to handle* ***multimodal inputs****, as often encountered in music-related applications (e.g., captioning or cross-modal summarization), remains an even more challenging yet crucial direction for future investigation.*

3.1.3 Other evaluation approaches. As previously mentioned, **human evaluation** is widely regarded as the gold standard for assessing open-ended text generation tasks and automatic metrics seek to correlate with human judgments. Human evaluation is also useful when automatic metrics have a harder time to capture more subjective or complex aspects of quality. Typical evaluation dimensions included in human studies of recommender systems are [135, 157]:

- **Fluency or grammaticality**, which measure how natural, readable, and error-free the output is;
- **Coherence or consistency**, which assesses whether the text flows logically without contradictions;

- **Relevance, adequacy, or informativeness**, which reflects how well the generated text preserves important content from the source or fulfills the task;
- **Veracity, factuality or faithfulness**, which examines whether the output is accurate and grounded in the source material or external knowledge (this is also a measure of hallucination presence);
- **Diversity, engagingness and insightfulness**, which captures originality and avoidance of repetition, as well as how interesting or likeable the output is (relevant in creative tasks such as music captioning).

Certain applications also employ task-specific criteria; for instance, **Coverage and conciseness** in summarization [78]. Quality judgments are often collected through Likert scales (e.g., 1–5 or 1–7) or pairwise preference rankings, in which annotators directly compare outputs. Significant efforts have also been made to assess the correlation between human judgments on these complex dimensions and existing automatic metrics [157], with mixed results. For example, **Coherence** has been shown to correlate relatively well with n-grams metrics metrics (e.g. BLEU, METEOR), whereas **Insightfulness** often shows no correlation [157].

However, as we mentioned before, given that the evaluation involving humans still requires significant efforts for collecting results on these subjective dimensions, more recent works employ **LLM-as-judges** [10, 24, 45, 163]. LLMs appear to be reliable evaluators in a large range of tasks. However, they also exhibit substantial inconsistency depending on the evaluated dimension (e.g., **engagingness** has low correlation with human judgement while **informativeness** higher), expertise level of human annotators (if compared with expert vs. non-expert human annotators), and whether the assessed text is human- or model-generated [10].

Therefore, LLMs should be rigorously validated against human judgments before being adopted as automatic evaluators. *In the context of music and music recommender systems, this raises an open and important question: to what extent can LLM-based evaluators reliably and fairly assess music-related textual content such as tags or generated descriptions, given the subjectivity and cultural variability inherent to musical preference and user perception? Addressing this question is required to ensure the reliability, bias control, and validity of LLM-driven evaluation in such personalized and domain-specific settings.*

3.1.4 Risks Discussion. Multiple studies from the NLP domain [22, 93, 107, 121, 121, 150, 153] have outlined other pitfalls in evaluating LLM-generated text, besides those mentioned before. We identify risks stemming from the model itself and its intricacies; related to the characteristics of the generated output; and associated with the evaluation metrics. We discuss further several of them, although, this list is not meant to be exhaustive.

Output-Related Risks. In generative tasks, where the output can take many possible forms, recent research stemming from the NLP field has identified several evaluation-related risks. First, the length of the generated text has been shown to strongly influence evaluation outcomes when using both n-gram–based and semantic similarity–based metrics [121]. *Because LLMs often generate longer texts than the ground-truth references, computing evaluation metrics on automatically summarized outputs—using simple extractive summarization methods—can, in practice, significantly improve the alignment of automatic metric scores with human judgments.* Second, Long et al. [93] show that all LLMs exhibit significant biases across various output formats (e.g., multiple-choice question formats, text wrapping styles, or list structures) spanning a wide range of benchmarks. *Several strategies to mitigate these biases are proposed, including (1) incorporating format demonstrations in the prompt and (2) repeating format instructions multiple times within the prompt.*

Metrics-Related Risks. Metric-related risks include social and representational biases, such as those observed with BERTScore [153]. Reference-free metrics [22] also pose challenges: they can favor models similar to themselves, and

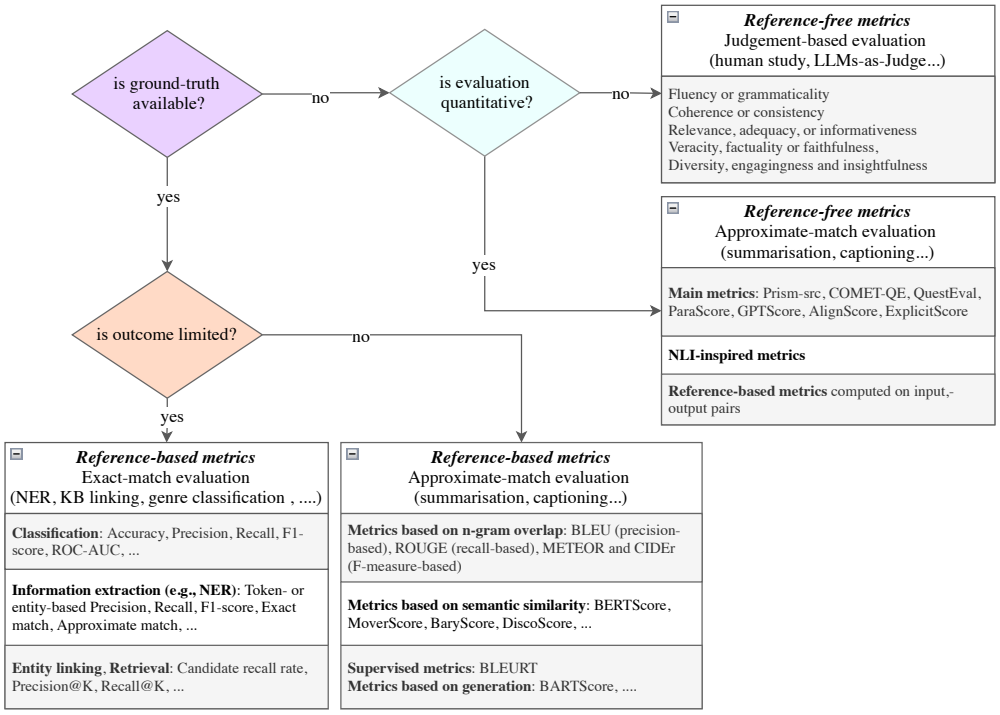

Fig. 4. NLP-driven evaluation framework: decision process and metrics overview.

thus, even penalize higher-quality human outputs, suggesting they are better suited as diagnostic tools rather than absolute measures of task performance. Moreover, when LLMs are employed as evaluators, they have been shown to exhibit position bias [163]. *That is, the order in which candidate responses appear in the prompt can skew evaluation results, potentially making one model appear significantly superior to another. To mitigate this bias, Wang et al. [163] propose a calibration framework, which also demonstrates closer alignment with human judgments.*

LLMs and Decoding Strategies. Given an input, the same LLM can produce highly variable outputs depending on the decoding strategy used (e.g., greedy decoding, nucleus sampling), a phenomenon referred to as non-determinism in generation [150]. However, for practical and computational reasons, LLM evaluations are often conducted using a single input–output pair. A recent study [150] evaluating multiple LLMs across several benchmarks has shown a significant gap between greedy decoding and sampling based decoding: greedy decoding often yields better results than the average performance obtained from multiple outputs generated through sampling strategies. Also, employing a best-of-N sampling strategy (selecting the best output among N generations for the same input) can enable even smaller LLMs to match or outperform larger models. *These findings carry important implications for the evaluation of LLMs. First, the inherent non-determinism of generation introduces a source of variability that challenges the reliability and reproducibility of evaluation results: a single input–output pair cannot adequately represent model performance. Second, decoding choices can significantly alter performance scores, making cross-model comparisons difficult unless decoding settings are standardized or explicitly reported. Third, methods such as best-of-N sampling reveal that a model's potential*

quality can be much higher than what single-sample evaluations suggest, raising questions about what aspect of performance (average, best-case, or expected quality) should be measured.

Fig. 4 summarizes the content discussed in this section, presenting it as a decision process to guide the selection of the most suitable evaluation metrics based on the specific characteristics of each scenario.

3.2 MRS-Driven Evaluation

As discussed before, the evaluation of LLMs for music recommendation is inherently multi-faceted because outputs must be *grounded* in a living catalog, *adhere* to nuanced natural-language constraints, promote *discovery* beyond the mainstreamness, and remain *controllable* and *robust* under different prompting regimes. Section 3.1 has reviewed NLP-driven evaluation strategies, metrics, and risks; further, we specialize evaluation to MRS. Concretely, we structure metrics along two complementary axes:

- **Measures of success**—what should improve when the system works well; and
- **Measures of risks**—failure modes that success metrics alone may mask.

Scope and assumptions. As discussed in Section 2.3, we consider the natural language recommendation task where a model produces a ranked list of k resolvable catalog items in response to an NL request, possibly conditioned on an NL user profile, and optionally having in-context examples, retrieved documents, or an explicit reasoning scaffold. While some evaluation dimensions, further presented, are general and applicable across domains, others are specifically designed to capture the unique characteristics and challenges of the music domain. Music-specific constraints (e.g., multilingual metadata, item aliasing / versioning, weekly new releases) motivate going beyond classical relevance to include grounding, coverage, and sequencing aspects. *Sequence-level* properties are especially salient for playlists (ordering, transitions), not only items in isolation.

3.2.1 Measures of Success. We define success as *faithfully satisfying a user's natural language query with real, resolvable catalog items*, while simultaneously promoting discovery and enabling controllable personalization. To this end, we propose success criteria along **six key dimensions** (G1–G6) outlined below.

G1. Query Adherence & Groundedness. It aims to evaluate whether the recommended items *satisfy the user's query constraints* and are *grounded* in real data (i.e., correspond to actual catalog items and factual knowledge). It ensures the recommendation answers the query faithfully *without* fabricating details [1, 27, 81]. With other words, do recommended items satisfy the stated constraints and exist as resolvable catalog entities[7]?

G2. Discovery Quality. This evaluation dimension captures the LLM's ability to surface *novel*, *diverse*, or *long-tail* content, essential for exploratory use cases. Specific metrics thus focus on how well recommendations introduce users to items they have not encountered before, span a broad range of content, and go beyond mainstream popularity [76, 159].

G3. Personalization Gain. It measures the improvement in recommendation utility attributable to incorporating user-specific information (e.g., a natural language profile), relative to a non-personalized or profile-agnostic baseline. This can be operationalized as the difference in a core utility metric or ranking objective, computed with and without profile conditioning. It often reflects how unique each user's recommendation list is relative to others or to popular trends, indicating the system's ability to tailor results to individual taste [8, 72].

G4. Profile Fidelity & Controllability. It assesses how well the LLM aligns its recommendations with the user's editable NL profile ("profile-as-text") and how effectively the LLM responds to explicit user edits of preferences

[7]One might verify items or items metadata against a knowledge base (entity resolution and linking) and penalize fabrications or misattributions.